AIDS2026: 9598

Leveraging integrated digital health systems to reduce community health worker reporting burden and strengthen differentiated HIV service delivery in Malaysia

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General
Category: E7: Health systems, health systems strengthening and partnerships
Country of research: Malaysia
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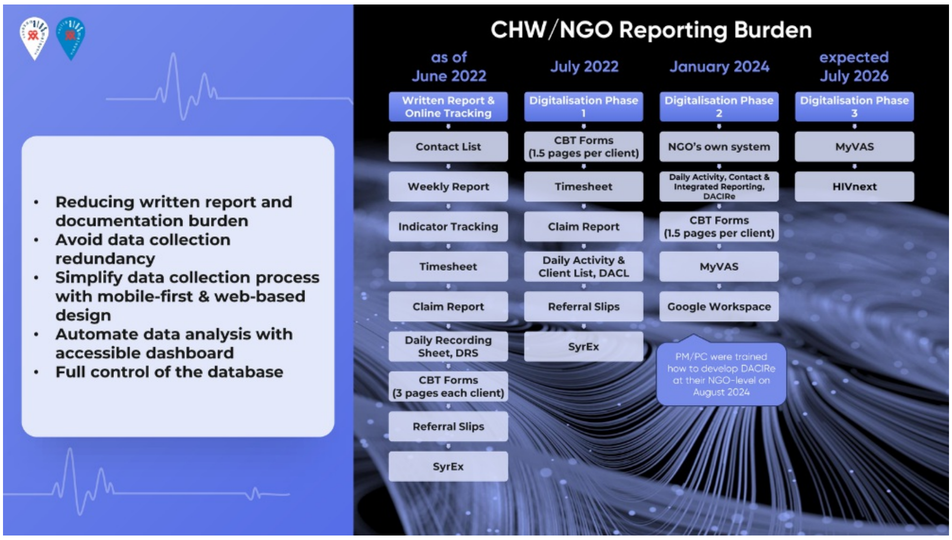
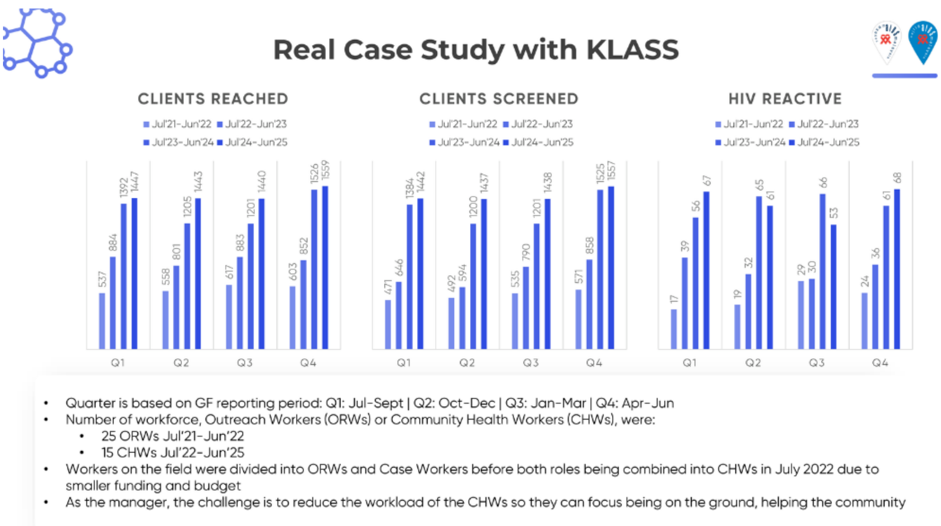
Abstract Text (max 350 words)
Background: Differentiated HIV Services for Key Populations (DHSKP) in Malaysia rely on Community Health Workers (CHWs) to serve high-risk groups. Previously, fragmented reporting using paper tools and standalone spreadsheets caused excessive administrative burden, data redundancy, and workforce burnout. This initiative aimed to strengthen the national response by implementing an integrated digital health information system to automate data management and improve operational efficiency at Kuala Lumpur AIDS Support Services (KLASS).
Description: We executed a phased digital transformation (2022–2025) to dismantle redundant workflows. Initially, CHWs navigated fragmented handwritten logs and manual entry into 'SyrEx' (a SQL-based app), resulting in high error rates. We first co-designed the DACL, a unified Excel-based tool consolidating five reporting streams. In 2023, KLASS upgraded this to ‘DACIRe’ within Google Workspace, introducing automation for claims and dashboards, which reduced reporting time by 50%. The initiative culminated in early 2024 with migration to MyVAS (Malaysia’s Virtualhealth Assistant System), a national legacy COVID-19 infrastructure. A dedicated DHSKP module now captures daily outreach with automated validation and national registry interoperability. While MyVAS is now the exclusive national monitoring platform, internal NGO systems remain heterogeneous; our focus was to set an example mechanism into the standardized national framework.



Lessons learned: Co-designing with CHWs ensured rapid adoption across a diverse workforce (aged 21–65). Centralization and mobile accessibility substantially improved data completeness, while real-time dashboards facilitated evidence-based decision-making. Crucially, digital efficiency proved vital for resilience. In July 2022, funding cuts forced KLASS to downsize its workforce by 48% (from 31 staffs to 15 polyvalent CHWs). Contrary to expectations of decline, the digital shift enabled this smaller team to increase direct outreach. Consequently, quarterly screenings rose by >200% (471 to 1,557) and HIV reactive case identification quadrupled (17 to 68) over three years (Q1Y1 vs Q4Y4).



Conclusions/Next steps: Integrated digital systems strengthen service delivery by minimizing reporting burdens and enabling real-time management. We are currently deploying a unified, non-SaaS ecosystem for all partner NGOs, expanding modules to include Community-Based Care, Inventory, and Staff Management and more modules as the program expands. This provides a scalable, cost-effective blueprint for digital transformation in resource-constrained settings.

Additional questions
Ethical research declaration: Yes

IAS digital learning platform
IAS+: Differentiated service delivery, Community leadership, HIV testing, Other
HIV Prevention Pre-conference: Yes

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